

Amphibians and Reptiles as Bioindicator Organisms: An Ecotoxicological and Ecological Review

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ABSTRACT

Amphibians and reptiles are increasingly4 recognized as highly effective bioindicator organisms for assessing environmental pollution and ecosystem health due to their unique physiological and ecological characteristics. This review provides a comprehensive and integrative evaluation of the biological, ecological, and methodological foundations underlying the use of herpetofauna in biomonitoring studies. Particular emphasis is placed on their dual habitat utilization, pollutant accumulation capacity, and sensitivity to both acute and chronic environmental stressors. The review synthesizes current knowledge on biomarker responses at multiple levels of biological organization, including biochemical, molecular, histopathological, and behavioral endpoints. The effects of major contaminant groups such as pesticides, heavy metals, endocrine-disrupting compounds, and emerging pollutants like microplastics are critically examined in relation to exposure pathways and species-specific responses. In addition, the role of amphibians as early-warning indicators and reptiles as long-term accumulation monitors is comparatively analyzed, highlighting their complementary value in integrated monitoring frameworks. Methodological approaches, including multi-biomarker strategies, genetic assays, and emerging omics-based techniques, are evaluated in terms of sensitivity, applicability, and limitations. The review also addresses key challenges in the field, such as the lack of standardization, variability among species and environments, and the need for long-term datasets. Finally, the study proposes a framework for the development of national biomonitoring programs with particular reference to the diverse and ecologically significant herpetofauna of Türkiye. By integrating species selection, sampling design, and biomarker application, this review aims to contribute to the advancement of ecotoxicological monitoring and to support evidence-based environmental management and conservation policies.

Biyoidikatör Organizmalar Olarak Amfibiler ve Sürüngenler: Ekotoksikolojik ve Ekolojik Bir Derleme

ÖZET

Amfibiler ve sürüngenler, kendilerine özgü fizyolojik ve ekolojik özellikleri sayesinde çevresel kirliliğin ve ekosistem sağlığının değerlendirilmesinde son derece etkili biyoidikatör organizmalar olarak giderek daha fazla önem kazanmaktadır. Bu derleme, herpetofaunanın biyoizleme çalışmalarındaki kullanımına ilişkin biyolojik, ekolojik ve metodolojik temelleri kapsamlı ve bütüncül bir yaklaşımla değerlendirmektedir. Özellikle bu organizmaların çift yaşam ortamına sahip olmaları, kirlетici biriktirme kapasiteleri ve hem akut hem de kronik çevresel stres faktörlerine karşı yüksek duyarlılıkları vurgulanmaktadır. Çalışmada, biyobelirteç yanıtları; biyokimyasal, moleküler, histopatolojik

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Article history

Received: 23/03/2026

Accepted: 19/04/2026

Available online: 22/06/2026

Keywords:

Bioindicator, amphibians,
reptiles, ecotoxicology,
biomarkers, environmental
monitoring, heavy metals,
pesticides

Anahtar Kelimeler:

Biyoidikatör, amfibiler,
sürüngenler, ekotoksikoloji,
biyobelirteçler, çevre izleme,
ağır metaller, pestisitler

How to Cite:

M. Kurnaz
"Amphibians and Reptiles as
Bioindicator Organisms: An
Ecotoxicological and Ecological
Review", *Environmental
Toxicology and Ecology*, c. 6,
sayı. 1, ss. 1-15, 2026.

DOI: 10.59838/etoxec.1914565

ve davranışsal düzeylerde ele alınarak çok katmanlı bir analiz sunulmuştur. Pestisitler, ağır metaller, endokrin bozucu maddeler ve mikroplastikler gibi başlıca kirlenici grupların etkileri; maruziyet yolları ve türlere özgü yanıtlar bağlamında kritik olarak incelenmiştir. Ayrıca amfibilerin erken uyarı sistemi olarak, sürüngeçerlerin ise uzun vadeli kirlenici birikim göstergesi olarak oynadığı roller karşılaştırmalı biçimde değerlendirilmiş ve bu iki grubun birlikte kullanımının önemi ortaya konmuştur. Biyobelirteç temelli yöntemler, genetik analizler ve yeni nesil omik yaklaşımlar; duyarlılık, uygulanabilirlik ve sınırlılıkları açısından ele alınmıştır. Bunun yanı sıra, alandaki temel sorunlar olan standardizasyon eksikliği, türler arası değişkenlik ve uzun dönemli veri ihtiyacı da tartışılmıştır. Son olarak, Türkiye'nin zengin herpetofaunası temel alınarak ulusal biyoizleme programlarının geliştirilmesine yönelik bir çerçeve önerilmiştir. Tür seçimi, örnekleme stratejileri ve biyobelirteç uygulamalarının entegrasyonu ile bu çalışma, ekotoksikolojik izleme çalışmalarının geliştirilmesine ve bilim temelli çevre yönetimi ile koruma politikalarına katkı sağlamayı amaçlamaktadır.

Anahtar Kelimeler: Biyoidikatör, amfibiler, sürüngeçerler, ekotoksikoloji, biyobelirteçler, çevre izleme, ağır metaller, pestisitler

1. INTRODUCTION

The accelerating pace of industrialization, urbanization, and intensive agricultural activity on a global scale is steadily increasing the pollutant burden discharged into ecosystems, placing severe and often irreversible pressure on biodiversity. Although physicochemical analytical methods can measure pollutant concentrations at a given point in time, this approach falls short in capturing the effects of long-term accumulation and the functional consequences for living systems [1]. In this context, bioindicator organisms have assumed an irreplaceable role as integrative measures of environmental change.

The concept of a bioindicator refers to the indirect assessment of environmental quality through the presence, absence, abundance, or physiological condition of a given organism [2]. An ideal bioindicator should have a wide geographic range, be relatively straightforward to sample, possess a sufficiently long lifespan to cover the period of exposure, and have the capacity to accumulate contaminants (Figure 1). In terms of satisfying these criteria simultaneously, herpetofauna, comprising amphibians (class Amphibia) and reptiles (class Reptilia), represents an exceptionally suitable biological group.

Amphibians are extraordinarily sensitive to environmental pollution owing to their thin, semi-permeable skin, their life stages spanning both aquatic and terrestrial environments, and their extended larval development [3]. Approximately forty per cent of amphibian populations worldwide are threatened, the highest proportion among all vertebrate groups; this figure is widely interpreted as a direct reflection of habitat deterioration [4, 5]. Reptiles, by contrast, offer unique tools for monitoring the chronic pollution of terrestrial ecosystems by virtue of their long lifespans, high accumulation capacities, and extensive home ranges [6, 7].

The present review addresses the following core questions: What biological characteristics make herpetofauna particularly suited as bioindicators? Which pollutants trigger which biomarker responses in these groups? How do the applicable methodologies differ between amphibians and reptiles? And how might national biomonitoring programs be designed within the context of the Turkish herpetofauna?

Key Selection Criteria for an Ideal Bioindicator Organism

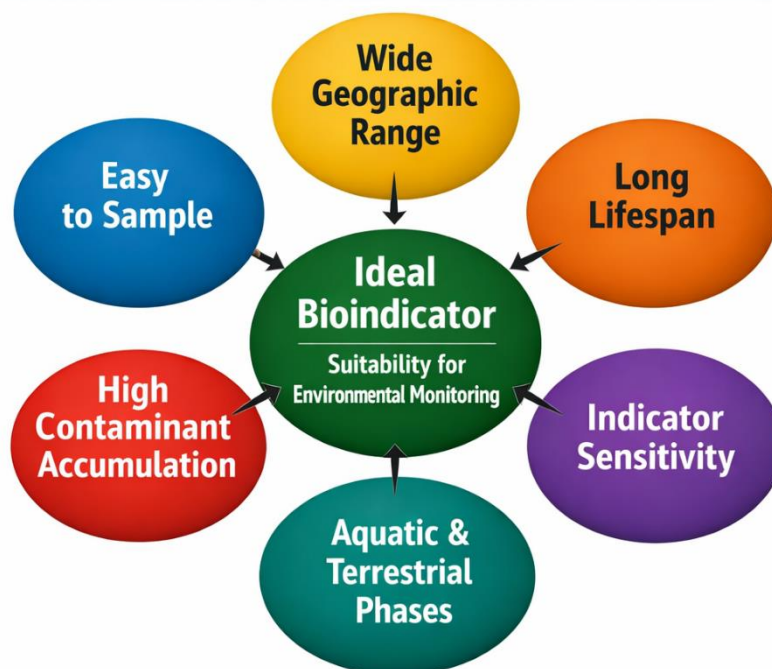


Figure 1. Key selection criteria for an ideal bioindicator organism. The simultaneous fulfilment of all six characteristics places herpetofauna in an exceptional position for environmental monitoring programs.

2. THE BIOINDICATOR CONCEPT AND CLASSIFICATION

2.1. Definitions and Historical Background

The term bioindicator was first employed in the early twentieth century by freshwater ecologists seeking to assess the organic pollution level of rivers using saprophytic microorganisms. Over time the concept expanded to terrestrial ecosystems, aquatic systems, and broader taxonomic groups [8]. According to current definitions, bioindicators are classified under three principal categories: (i) presence–absence indicators, (ii) sensitivity indicators, and (iii) accumulation indicators [9]. Herpetofauna is represented across all three categories but stands out particularly in its roles as a sensitivity and accumulation indicator. Spellerberg [10] described herpetofauna as the "silent witnesses of ecosystem health".

2.2. The Biomarker Concept and its Levels

A biomarker is defined as a measurable biological parameter in an organism that changes in response to an environmental stressor. Van der Oost et al. [11] classified biomarkers at three principal levels: exposure biomarkers, effect biomarkers, and susceptibility biomarkers (Figure 4). Exposure biomarkers directly measure the pollutant burden in an organism. Effect biomarkers reveal whether the pollutant has produced biological damage. Susceptibility biomarkers explain why individuals respond differently to pollutants due to genetic variation [12, 13].

3. AMPHIBIANS AS BIOINDICATOR ORGANISMS

3.1. Biological Characteristics and Sensitivity Mechanisms

The bioindicator value of amphibians derives primarily from their life cycle bridging terrestrial and aquatic environments (Figure 2). Most species spend the early stages of their lives in water, during which they are directly exposed to pollutants dissolved in the water column through their thin, vascularized, and moist skin [14, 15]. This permeable skin structure places amphibians in a position of high sensitivity to aquatic pollution, whilst simultaneously providing a route for the epidermal uptake of atmospheric contaminants and pesticides during the terrestrial stage.

Amphibian eggs lack a protective calcium shell; the gelatinous envelope constitutes an inadequate barrier against contaminants present in the aquatic environment. The embryonic development stage therefore represents a critically sensitive window of exposure: morphological malformations (particularly limb deformities), developmental delay, and embryonic mortality rate serve as sensitive indicators of pollutant stress [16, 17].

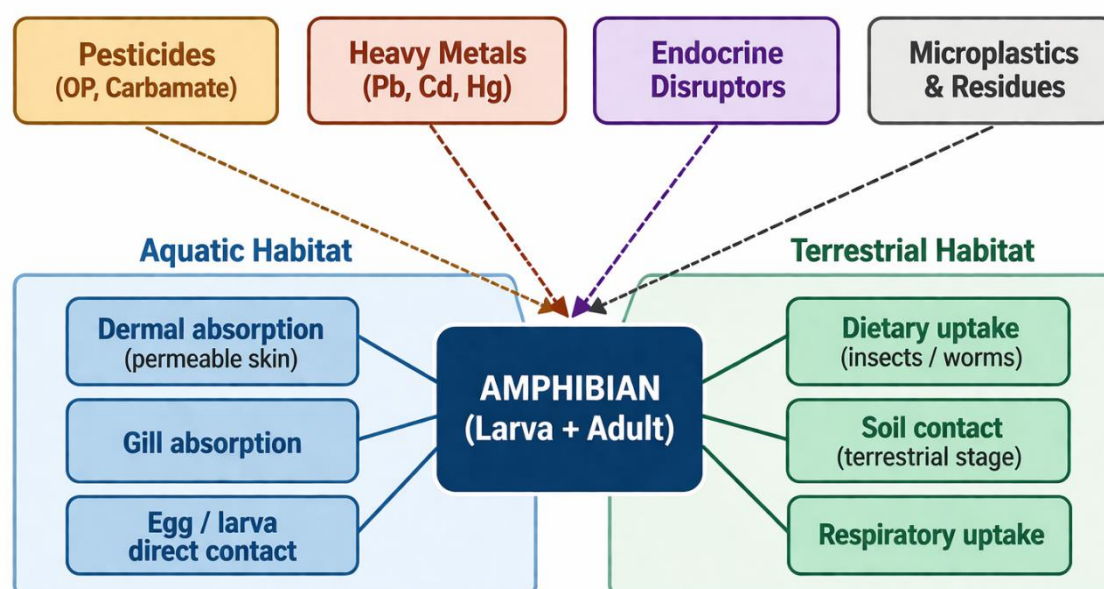


Figure 2. Environmental pollutant exposure routes in amphibians. Different exposure pathways operate simultaneously in the aquatic environment (larval stage) and the terrestrial environment (adult stage), making amphibians integrated bioindicators of dual-habitat pollution.

3.2. Response to Pesticides: Nervous System and Endocrine Disruption

Organophosphate and carbamate pesticides exert neurotoxic effects on amphibians by inhibiting the enzyme acetylcholinesterase (AChE). In larvae of *Rana temporaria* exposed to sublethal concentrations of common agrochemicals such as deltamethrin, chlorpyrifos, and imidacloprid, AChE activity has been reported to be suppressed by between thirty and seventy per cent [18, 19]. Glyphosate-based herbicides have additionally been shown to exert immunosuppressive effects in amphibians, thereby increasing susceptibility to secondary infections [20].

Hayes et al. [21] demonstrated that concentrations of atrazine as low as 0.1 ppb are sufficient to induce gonadal feminization in male *Xenopus laevis*. This seminal finding clearly established the capacity of amphibian biomonitoring to detect endocrine-disrupting compounds at concentrations far below the thresholds detectable by standard ecotoxicological tests [22].

3.3. Heavy Metal Accumulation and Oxidative Stress

Heavy metals such as lead (Pb), cadmium (Cd), mercury (Hg), and zinc (Zn) tend to accumulate in amphibian tissues, particularly the liver and kidney. In *Pelophylax ridibundus* populations inhabiting contaminated catchments, liver Cd concentrations have been found to be between six and twelve times higher than those in reference populations. Antioxidant enzyme activities including superoxide dismutase (SOD), catalase (CAT), and glutathione S-transferase (GST) and malondialdehyde (MDA) levels are reliably induced at concentrations well below lethal doses, offering superior sensitivity for the detection of sublethal chronic exposure conditions [23].

3.4. Morphological Malformations, Population-Level Effects, and Disease Dynamics

Strong evidence supports the view that amphibian morphological malformations arise from the synergistic interaction of trematode parasites, endocrine-disrupting chemicals, and UV-B radiation [24]. Disrupted sex ratios, reduced reproductive success, and declining species abundance constitute population-level parameters that can be integrated into long-term biomonitoring programs [17]. Furthermore, the fungal pathogens *Batrachochytrium dendrobatidis* (*Bd*) and *B. salamandrivorans* (*Bsal*) have become a major component of the global amphibian crisis, causing catastrophic mortality particularly in populations already under pollution and climate stress [5].

Table 1. Amphibian Species Prominent in Bioindicator Research

Species	Taxon	Bioindicator Property	Monitored Parameter	References
<i>Rana temporaria</i>	Anura / Ranidae	Early developmental disorder, malformation	Pesticides, heavy metal accumulation	[25]
<i>Pelophylax ridibundus</i>	Anura / Ranidae	Liver enzyme activity, histopathology	Water quality, endocrine disruptors	[25]
<i>Bufo bufo</i>	Anura / Bufonidae	Egg and larval sensitivity	UV radiation, pH change	[26]
<i>Triturus cristatus</i>	Caudata / Salamandridae	Habitat quality, population density	Wetland degradation, pollution	[27]
<i>Salamandra salamandra</i>	Caudata / Salamandridae	Skin secretion alterations	Fungal disease (<i>Batrachochytrium</i>)	[28]
<i>Hyla arborea</i>	Anura / Hylidae	Calling activity phenology	Climate change, habitat monitoring	[29]
<i>Xenopus laevis</i>	Anura / Pipidae	Gonadal feminization, intersex	Atrazine, endocrine disruptors	[30]
<i>Bufotes viridis</i>	Anura / Bufonidae	Reproductive phenology shift	Climate change, urban heat	[31]

4. REPTILES AS BIOINDICATOR ORGANISMS

4.1. Advantages and Methodological Foundations

Long lifespan, dependence on terrestrial habitats, and a generally elevated position in the food chain make reptiles a distinct and complementary tool to amphibians for monitoring chronic and biomagnified pollutants [6, 7]. The relatively small home ranges of many lizard and snake species allow pollutant load values to be geographically assigned, enabling biogeographic monitoring applications. The movement of pollutants through the food chain, **biomagnification**, is schematized for reptiles in Figure 3.

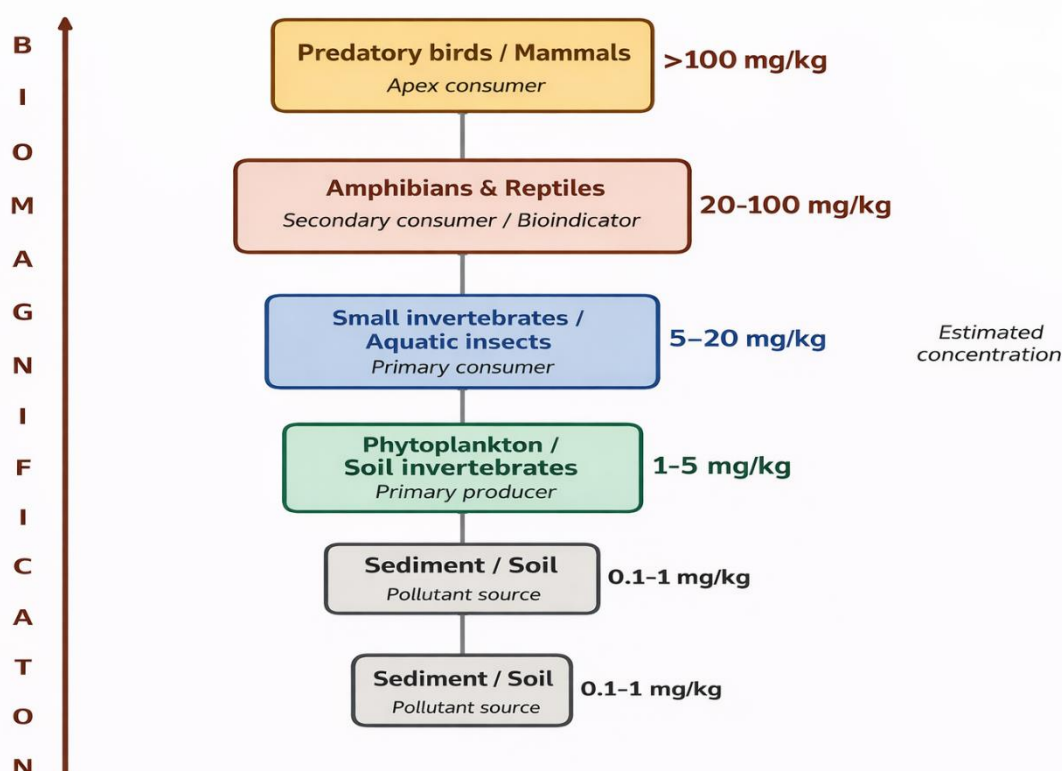


Figure 3. Food chain biomagnification in terrestrial and aquatic ecosystems and the position of amphibians and reptiles. Estimated concentration values are conceptual and subject to considerable variation by species and pollutant.

4.2. Pollutant Accumulation in Lizards and Snakes

Lacerta agilis individuals sampled from pesticide-treated agricultural land have been shown to carry elevated concentrations of organochlorine pesticides and heavy metals in liver, kidney, and adipose tissues [32]. In *Natrix natrix* and *Nerodia sipedon*, methyl mercury concentrations and PCB levels have been reported to reach values ten to one hundred times higher than those in primary consumer controls [33]. In urban environments, studies on *Podarcis muralis* have detected markedly elevated DNA damage levels, reduced fertility, and altered thermoregulatory behavior patterns in individuals inhabiting habitats close to impervious surfaces [32, 33, 34].

4.3. Long-Term Accumulation Monitoring in Chelonians

Chelonians are regarded as unique biological archives integrating chronic pollutant exposure over decades, by virtue of their long lifespans and the high accumulation capacity of their lipid-rich tissues [6]. *Testudo graeca* and *Emys orbicularis* are recommended as candidate monitoring species for soil and water pollutants across Türkiye. Liver cadmium concentrations in *Caretta caretta* individuals have been shown to increase linearly with age, functioning as an objective measure of long-term Mediterranean pollution levels [35, 36].

4.4. Genotoxicity and Cancer Biomarkers

The Comet assay and micronucleus test have become standard procedures for assessing pollutant genotoxicity in various lizard and chelonian species. In *Podarcis muralis* individuals inhabiting habitats near industrial pollution sources, significantly elevated DNA damage levels compared to reference populations have been detected [34]. Liver histopathology is widely used in reptile ecotoxicology as a macroscopic and microscopic indicator of chronic chemical stress [37, 38].

Table 2. Reptile Species Prominent in Bioindicator Research

Species	Taxon	Bioindicator Property	Monitored Parameter	References
<i>Lacerta agilis</i>	Squamata / Lacertidae	Tissue and blood biomarkers	Heavy metal accumulation, insecticides	[32]
<i>Podarcis muralis</i>	Squamata / Lacertidae	Thermoregulatory behaviour change	Urban heat island effect	[34]
<i>Natrix natrix</i>	Squamata / Colubridae	Food chain accumulation monitoring	Methyl mercury, PCB accumulation	[33]
<i>Testudo graeca</i>	Testudines / Testudinidae	Long-lived accumulation monitor	Soil heavy metals, lead	[39]
<i>Emys orbicularis</i>	Testudines / Emydidae	Aquatic habitat integrity	Water pollutants, DDT residues	[40]
<i>Caretta caretta</i>	Testudines / Cheloniidae	Mediterranean chronic pollution archive	Cd, Hg, PCB, DDT metabolites	[41]
<i>Ablepharus kitaibelii</i>	Squamata / Scincidae	Soil pollutant sentinel	Soil insecticides, agricultural habitats	[42]
<i>Darevskia sp.</i>	Squamata / Lacertidae	Endemic rock lizard sentinel	Soil pesticides, Anatolian endemism	[37]

5. APPLIED BIOMARKER METHODOLOGIES

5.1. Biochemical Biomarkers

Cholinesterase activity, particularly brain and plasma AChE, is the primary biochemical indicator of organophosphate and carbamate exposure. Oxidative stress biomarkers (SOD, CAT, GST, MDA, carbonyl proteins) are routinely applied in both amphibians and reptiles. Metallothionein expression has become an

increasingly used biomarker evaluated as a specific molecular response to heavy metal accumulation [23, 43]. Vitellogenin (VTG) is accepted as the gold standard for endocrine disruptor assessment because it is induced in male amphibians exposed to oestrogenic chemicals [22].

5.2. Genetic and Histopathological Methods

The Comet assay, micronucleus test, and sister chromatid exchange analysis have found widespread application in documenting the effects of genotoxic pollutants on herpetofauna. Histopathology remains indispensable for documenting the organ-level consequences of chronic pollutant pressure. Degenerative changes, foci of necrosis, and tumor formation observed in liver, kidney, and spleen tissues are evaluated using standardized morphological scoring systems.

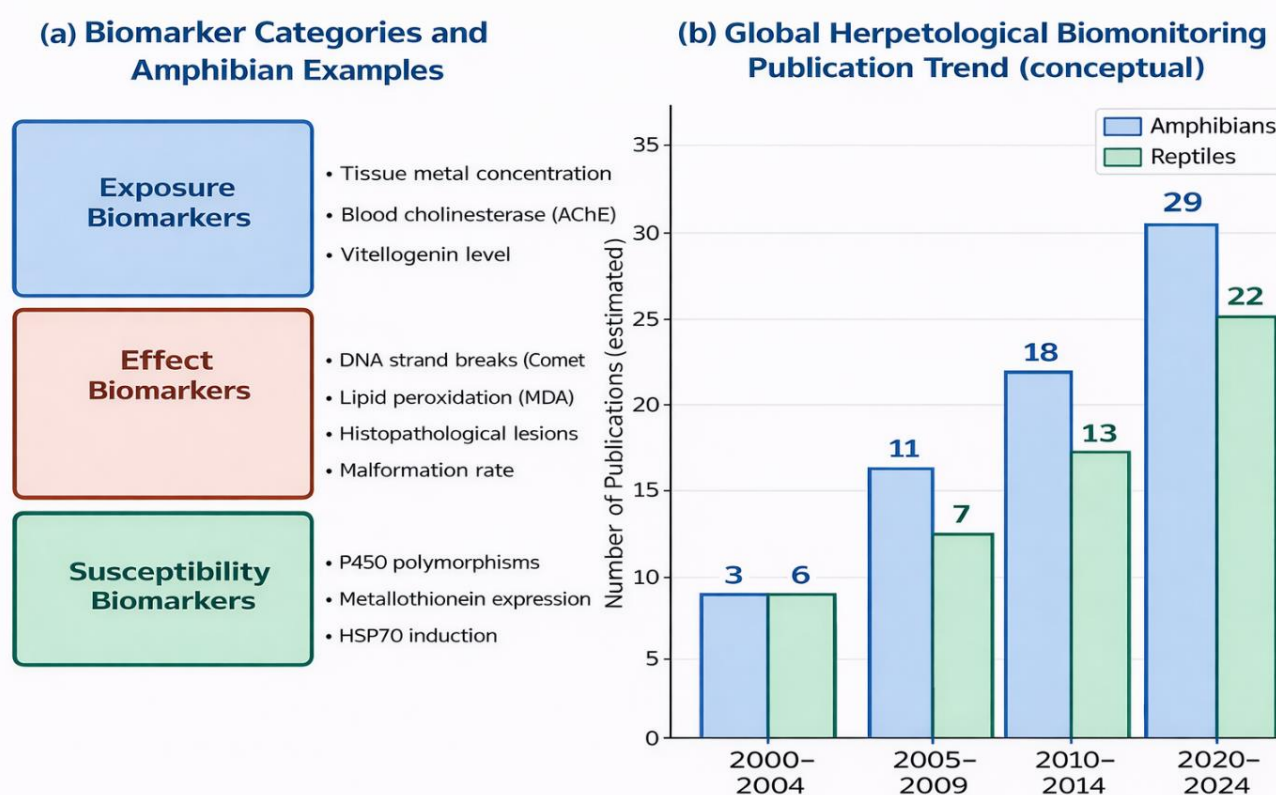


Figure 4. (a) Biomarker categories and methods applied to amphibians; (b) Global trend in herpetological biomonitoring publications (conceptual, estimated data). The growth in the literature is observed to be faster for amphibians than for reptiles.

5.3. Omics Approaches: Future Perspectives

The integration of transcriptomic, metabolomic, and proteomic analyses into field ecotoxicology represents the most promising frontier of next-generation herpetological biomonitoring. Gene expression profiles can illuminate sub-sublethal exposure scenarios and the interactions of complex pollutant mixtures that classical single-biomarker approaches are unable to detect [17]. HSP70 expression profiles associated with the stress response have been shown to exhibit a strong linear relationship with pollutant load in certain amphibian species [26, 37].

6. AMPHIBIANS AND REPTILES: COMPARATIVE BIOINDICATOR ANALYSIS

Although both groups possess strong bioindicator characteristics, they differ markedly in methodological, ecological, and biological terms. Amphibians serve as early warning bioindicators owing to their high sensitivity to acute pollutant stress, whilst reptiles, through their longer lifespans and greater accumulation capacity, maintain a biological record of chronic pollution history. These complementary strengths render multi-species approaches significantly more informative than single-species studies [7].

Table 3. Comparative Bioindicator Characteristics of Amphibians and Reptiles

Characteristic	Amphibians	Reptiles
Exposure route	Dermal absorption, gill, oral	Oral, respiratory, dermal (low)
Accumulation capacity	Moderate; acute exposure indicator	High; chronic and long-term monitor
Sensitivity	Very high; early warning system	Moderate–high; more tolerant
Biomarkers applied	AChE, GST, MDA, histopathology, malformation	SOD, catalase, DNA damage, liver tumour
Sampling ease	High; especially during larval stage	Moderate; some species difficult to catch
Monitoring ecosystem	Aquatic + terrestrial transition zones	Terrestrial; some species at aquatic edges
Example Turkish species	<i>P. ridibundus</i> , <i>B. viridis</i> , <i>H. arborea</i> , <i>T. cristatus</i>	<i>L. agilis</i> , <i>T. graeca</i> , <i>E. orbicularis</i> , <i>N. natrix</i>

The most appropriate strategy in monitoring program design is a multi-species approach that exploits the complementary strengths of both groups. In an agricultural catchment, for instance, amphibian larvae could be used to detect acute pesticide stress whilst lizard or snake populations from the same area are consulted to assess long-term organochlorine accumulation [1].

7. TURKISH HERPETOFAUNA AND NATIONAL BIOMONITORING POTENTIAL

7.1. Herpetological Diversity and Biogeographical Setting

Türkiye is situated at the intersection of the European, western Asian, and North African biogeographical zones, and hosts one of the most diverse herpetofaunas in the Western Palearctic. Approximately 180 amphibian and reptile species have been recorded across the country, a significant proportion of which are endemic or near-endemic to Anatolia [44, 45]. This extraordinary richness is a direct product of the country's unique position as a biogeographical crossroads, its varied topography, and the persistence of multiple Pleistocene refugia within its borders. Amphibian diversity is particularly high in the humid coastal zones of the Black Sea and Aegean–Mediterranean regions, where species such as *Pelophylax ridibundus*, *Bufotes viridis*, *Rana macrocnemis*, and *Pelodytes caucasicus* co-occur and serve as ecotoxicological sentinels in agricultural landscapes [42, 46, 47]. Among chelonians, *Mauremys rivulata* and *Emys orbicularis* are associated with freshwater habitats under intense agricultural and urban pressure, and have been used as bioindicators of water quality along Çanakkale waterways [37, 48, 49, 50]. The loggerhead sea turtle *Caretta caretta* is among the most important indicator organisms of Mediterranean coastal pollution; studies conducted on Turkish coasts have documented bioaccumulation of cadmium, mercury, lead, and persistent organic pollutants across different life stages and sexes [51, 52]. Among lizards, the

endemic rock lizards of the genus *Darevskia* offer rare opportunities for genotoxicity monitoring in soil-polluted habitats [37]. *Ophisops elegans*, another widely distributed Anatolian lacertid, has been used in controlled laboratory exposures to assess histopathological damage from carbamate insecticides [42].

Despite this exceptional biological wealth, the herpetological literature from Türkiye reveals a growing but still fragmented ecotoxicological knowledge base. Intense pesticide uses in the agricultural sector, industrialized river basins, and rapid urbanization are generating measurable pollutant loads across the country's ecosystems. Genotoxicological studies on *Pelophylax ridibundus* and *Bufo viridis* populations from the Çan and Biga streams in Çanakkale have detected elevated micronuclei frequencies and nuclear abnormalities attributable to industrial and agricultural effluents [38, 53]. Similarly, experimental exposures of native tadpoles of Black Sea amphibian species to agricultural fertilizer concentrations demonstrate ecologically relevant sublethal effects [47]. Ecotoxicogenomic frameworks and enzyme/hormone-level investigations in *Testudo graeca* and similar chelonians further underscore the depth of methodological approaches available for national monitoring [50, 54, 55]. Nevertheless, systematic, long-term, and multi-species biomonitoring programs targeting Türkiye's herpetofauna remain at an early stage compared with the international literature. Establishing such programs with standardized protocols, archival tissue banks, and multi-matrix sampling represents both the most pressing gap and the greatest opportunity in the field.

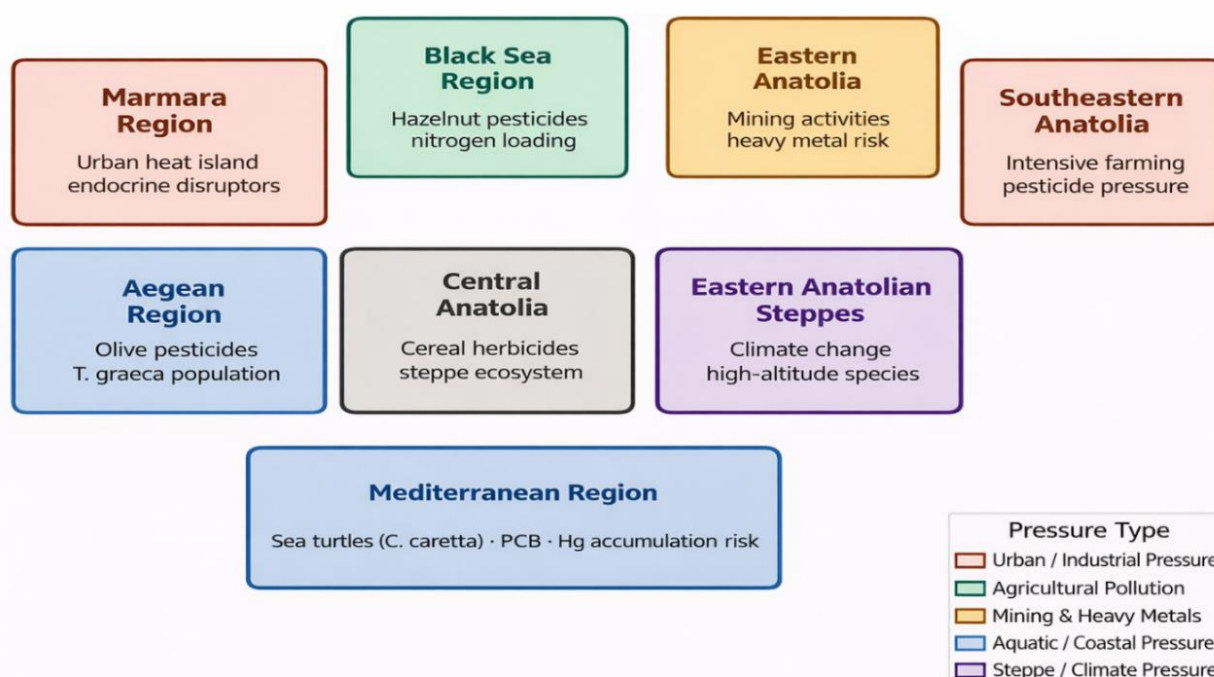


Figure 5. Schematic herpetofauna biomonitoring priority map of Türkiye by region and dominant pressure types. Each region differs with respect to the prevailing pollutant pressure type and the recommended monitoring species.

7.2. Priority Species and Methodological Recommendations

Among frogs, *Pelophylax ridibundus* and *Bufo viridis* stand out for pollution monitoring in agricultural areas and delta habitats. *Triturus cristatus* is considered a critical wetland quality indicator among salamanders. Among lizards, *Lacerta agilis*, *Ablepharus kitaibelii*, and various *Darevskia* species are primary candidates for terrestrial ecosystem monitoring. Among chelonians, *Testudo graeca* and *Emys orbicularis* are valuable monitoring species for soil and aquatic pollutants, especially in the Aegean, Mediterranean, and south-eastern Anatolian basins [39, 40].

The following methodological principles are recommended: (i) parallel sampling of reference and contaminated populations; (ii) a multi-matrix approach (biological sample + environmental matrix); (iii) a seasonal sampling schedule; and (iv) establishment of archival tissue banks. Implementing these principles would strengthen Türkiye's biotic monitoring component in compliance with EU environmental directives and provide internationally comparable data.

8. DISCUSSION

The broad literature evaluated in this review strongly supports the view that herpetofauna holds an irreplaceable role in environmental monitoring programs. The early warning system function of amphibians and the chronic accumulation archive character of reptiles should be placed at the core of monitoring program design, exploiting the complementary strengths of both groups.

The primary methodological challenge is the lack of standardization across studies. Differences in tissue types, age classifications, seasonal sampling times, and biomarker selection limit the scope for direct comparison of findings [13]. The development of protocol guidelines analogous to those of the European Committee for Standardization would go a long way towards addressing this problem.

The control of confounding factors, individual age, sex, seasonal physiology, dietary history, and habitat type must be addressed through adequate sample sizes and multivariate statistical approaches. Future priorities include the integration of omics technologies, investigation of microbiome–pollutant interactions, and long-term population studies focusing on the synergism between climate change and pollution.

9. CONCLUSION

This review has systematically demonstrated the unique and complementary bioindicator value that amphibians and reptiles offer in environmental monitoring program. Amphibians serve as early warning systems thanks to their high sensitivity to acute stress, whilst reptiles carry the biological record of chronic pollution history through their long lifespans and strong accumulation capacity.

Türkiye, with its extraordinary herpetological diversity and habitats facing diverse pollution pressures, offers an exceptional context for the establishment of comprehensive and sustainable national monitoring programs. It is hoped that this review will contribute to the conceptual framework of the field, inform discussions on methodological standardization, and provide a foundation for the design of original research studies, particularly for researchers working in Türkiye.

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